
Create a Redis Cluster Using Amazon ElastiCache

Prerequisites

Before you start:

- You must have an AWS account.
 - You should have an existing **VPC**, **subnets**, and **security groups**.
 - Your Redis clients (EC2/Lambda) must be in the **same VPC** and **allowed in SG**.
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Step-by-Step Guide (Using Console)

Go to ElastiCache Console

- Open AWS Console → **ElastiCache**
 - Choose **Redis** from the left panel
 - Click **Create Redis cluster**
-

Choose Cluster Mode

- ♦ Choose **Cluster mode disabled** for:
 - Single-node or primary + replicas

- Simpler setup (recommended for most apps)

♦ Choose **Cluster mode enabled** for:

- Redis **sharding**
- Advanced use cases (e.g., very large datasets)

For this guide, select "**Cluster mode disabled**"

3 Configure Cluster Details

Setting	Recommended
Name	my-redis-cluster
Engine version	Latest compatible Redis version
Port	6379 (default Redis port)

4 Choose Node Type and Capacity

Option	Value
Node type	cache.t3.micro (for dev/test)
Number of replicas	0 (or 1+ for high availability)

5 Network Settings

Option	Value
VPC	Choose your VPC
Subnet group	Choose or create a Redis Subnet Group

Security group Must allow access on port **6379** from your app/Lambda

💡 You can create a subnet group from **ElastiCache** → **Subnet groups**

6 Encryption & Auth

Option	Recommendation
In-transit encryption	✅ Enable
At-rest encryption	✅ Enable
Redis AUTH token	✅ Set one (save it securely)

7 Maintenance and Backups

- Enable automatic snapshots if needed
 - Choose a preferred maintenance window
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8 Review & Create

- Review all settings
- Click **Create**

🕒 Wait 2–5 minutes for the cluster to become **Available**

🔍 **After Creation**

✓ Get Endpoint

1. Go to **ElastiCache** → **Redis** → **your cluster**
2. Copy the **Primary Endpoint** (without port)

Example:

my-redis-cluster.xyz.cache.amazonaws.com

✓ Connect from EC2 or Lambda

EC2 (in same VPC):

```
sudo apt update
```

```
sudo apt install redis-tools -y
```

```
redis-cli -h <your-redis-endpoint> -a <your-auth-token>
```

Sample Command:

```
SET session:1 "user123"
```

```
GET session:1
```

Monitoring

Use **CloudWatch** to monitor:

- **Cache Hits / Misses**
- **Memory Usage**
- **CPU Utilization**
- **Evictions**

Clean Up (Optional)

To avoid charges:

- Go to ElastiCache → Redis → Select your cluster → Click **Delete**

Summary

Step	Action
1	Open ElastiCache console
2	Choose Redis and Cluster mode
3	Configure engine, node type
4	Select VPC, subnet, SG
5	Enable encryption & auth
6	Launch and connect via CLI or code
